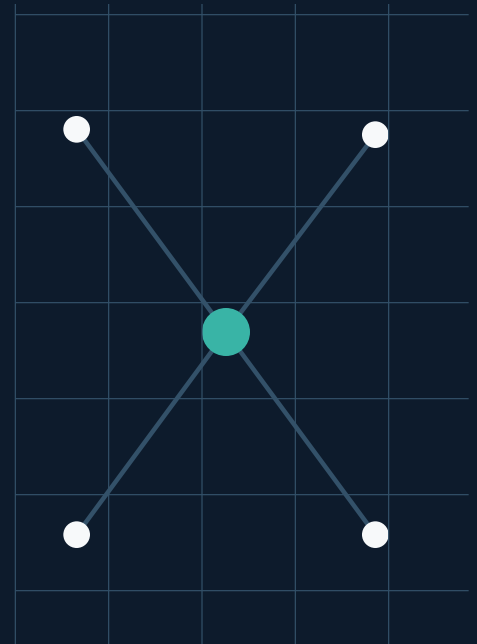


GRIDRESOLVE AI



Evidence before answers.

A governed, evidence-first multi-agent utility billing investigation system built on Microsoft Foundry.

POST-REVIEW VALIDATION

Five new hosted executions on workflow v11. A verified no-follow-up route, a successful reference-case regression, expanded evaluation coverage and independent communication controls.

09

SPECIALISTS

08

HOSTED RUNS

05

CASES

1,544

LOCAL CHECKS

02 / BUSINESS CHALLENGE

One complaint. Five decisions.

A confident allegation is an input to investigate, not a fact to adopt.

01 / BILLING

Do energy charges, rates and billed consumption reconcile?

02 / USAGE

What changed, and what is actually observed?

03 / EQUIPMENT

Do register reads or diagnostics establish a meter fault?

04 / POLICY

Which explanations and actions are allowed by governed rules?

05 / AUTHORITY

Who may approve an inspection, adjustment or closure? A customer-facing draft cannot authorize itself.

REFERENCE SYNTHETIC CASE / SYN-CASE-4003

A customer reports a sharply higher bill and asks the system to confirm that the meter failed and reverse the charge.

Design response: reconcile records, preserve uncertainty, release only a supported message, and retain financial authority with the human Billing Supervisor role.

03 / SYSTEM ARCHITECTURE

Nine agents. Distinct authority.

One hosted Foundry workflow, explicit task messages, shared case context and structured outputs.

INVESTIGATION / FOUR AGENTS

01 CaseTriageAgent

Classifies the request and material assertions.

02 AccountEvidenceAgent

Creates a cited ledger from case records.

03 UsageAnomalyAgent

Compares consumption and register movement.

04 PolicyKnowledgeAgent

Maps the relevant policies and constraints.

SYNTHESIS / TWO AGENTS

05 ResolutionPlannerAgent

Builds the claim ledger, proposed plan and follow-up decision.

06 CustomerCommunicationAgent

Drafts six human-readable customer response fields.

INDEPENDENT CONTROL / THREE AGENTS

07 EvidenceComplianceAgent v7

Reviews the message and issues an exact route token.

08 EscalationCoordinatorAgent

Prepares a package for the human decision role.

09 CaseAuditAgent

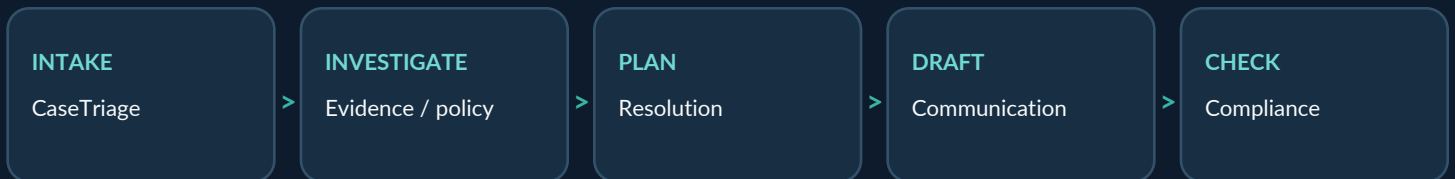
Records the observed execution route and provenance at termination.

Separation of duties: the drafter never approves its own answer; communication release and case follow-up are independently routed.

04 / FOUNDRY ORCHESTRATION

Governed workflow, version 11.

Published with EvidenceComplianceAgent v7; both definitions were read back from the hosted project.



EXACT, DETERMINISTIC RELEASE CONDITIONS

A structured **APPROVE** decision and exact route token

Complete investigation and six readable message fields

Only the most recently approved draft can be sent

CORRECTION LOGIC / LOCAL ENGINE VALIDATION

REJECT_REWRITE

Return to communication, regenerate the draft and recheck compliance.

REJECT_REPLAN

Return to planning, redraft the message and recheck compliance.

GUARDS / TWO ATTEMPTS MAXIMUM

Correction attempts are explicitly unrolled; no model controls the retry count. A malformed decision or further rejection takes the fail-closed route. Communication approval and human follow-up remain separate gates; audit is last.

The correction branches passed 76 local declarative-engine checks. The five hosted v11 cases produced compliant drafts at first review, so no correction action was invoked.

05 / REFERENCE CASE

The allegation met the records.

SYN-CASE-4003: genuine hosted workflow runs on v10 and v11; criteria fixed before execution.

PREVIOUS BILL

\$152.80

640 kWh

CURRENT BILL

\$203.40

870 kWh

REGISTER CHANGE

870 kWh

42,690 - 41,820

THE EVIDENCE VERDICT

The meter register movement equals the billed July consumption. The supplied diagnostics and event logs do not establish that the meter over-recorded energy.

The approved customer explanation does not treat the allegation as fact and does not promise a credit.

INDEPENDENT, OBSERVED OUTCOME

MESSAGE APPROVED

EvidenceComplianceAgent issued the reasoned APPROVE decision with zero failed checks.

HUMAN AUTHORITY PRESERVED

The separate follow-up gate prepared a Billing Supervisor review package.

14 / 14

Pre-registered hosted acceptance criteria passed on v10 and again on v11, with the same release and follow-up route.

06 / HOSTED VALIDATION

Eight executions. Five synthetic cases.

Three preserved historical runs plus five new hosted v11 runs on 23 September 2026.

FIVE NEW V11 RUNS

Every new run produced an APPROVE decision and a message without a meter-fault conclusion or credit promise. Four retained human follow-up; one closed without it.

CASE	TARGET	ACCEPTANCE	OBSERVED
4007	Reasoned escalation	8 P / 5 F / 1 N	Approved + review
4001	No follow-up	12 / 12 PASS	Closed
4003	Reference regression	14 / 14 PASS	Approved + review
4011	Rewrite path	8 P / 1 F / 5 N	Approved + review
4002	Replan path	9 P / 1 F / 5 N	Approved + review

P = PASS / F = FAIL / N = NOT OBSERVABLE

WHAT THE PLATFORM ACTUALLY RECORDED

The no-follow-up branch was observed for the first time on 4001. The 4003 regression passed. The conflicting-record case released a cautious message and independently preserved human review. Rewrite, replan and non-approval reason codes were exercised locally, not triggered by these hosted runs.

07 / REVIEW RESPONSE

Feedback translated into engineering.

The revised contract, evaluation scheme and evidence archive are published alongside the original run history.

EXPLAINABLE ESCALATION

Compliance v7 defines cited reason codes on every non-approval. The contract passed 21 local checks; no v7 hosted rejection occurred.

BOUNDED CORRECTION

Workflow v11 implements separate rewrite and replan paths, limited to two attempts. Passed 76 local workflow checks.

GROUNDING ROOT-CAUSE LABELS

Ground truth v2.1 separates claim verdict from root cause; historic scores remain frozen and newly discovered label disagreements are recorded.

MORE TARGETED COVERAGE

36 evaluation cases and 18 adversarial probes, mapped by issue category and observed execution status.

HUMAN FEEDBACK RECORD

Append-only, reviewer-only review record distinguishes actual decisions from local simulation. Metrics return NotMeasured without real input.

INSPECTABLE EVIDENCE

Two annotated traces, per-run results, branch action IDs, acceptance plans, hash manifests and redacted community artifacts.

08 / EVALUATION

A pass names its test and scope.

Local, hosted, deterministic, platform and human outcome measurements are reported separately.

1,544

LOCAL PASSES

105 / 105

POWER FX CASES

11 / 11

FRAMEWORK PARITY

TWELVE LOCAL SUITES / 0 FAILURES

Routing and case data	83	Synthetic data integrity	151
Foundry runner	359	Evaluation package	198
Integration contracts	138	Reason-code contracts	21
Scenario cases	75	Supervisor feedback	62
Evaluation v2.1	125	Workflow engine v10	91
Workflow engine v11	76	Control Center	165

Evaluation discipline: each hosted case had hashed, pre-registered acceptance criteria. Branches not taken are NOT OBSERVABLE, not PASS. Historic v2.0 results are retained; corrected claim/root-cause labels use versioned v2.1 ground truth.

09 / OBSERVABILITY AND COST

The audit follows platform evidence.

Native traces, action IDs and usage blocks anchor the reported results.

EIGHT HOSTED RUNS

8

5 distinct synthetic cases

V11 WALL CLOCK

165-220 s

range across 5 new runs

TOKEN-BASED MODEL COST

\$0.3280

Five new v11 runs. Estimated from 465,202 input and 107,178 output tokens returned by Foundry.

Eight-run estimate: \$0.4591. Azure confirmed \$0.1310 billed for the original three; the five September 23 charges were not yet visible at the reporting time.

TRACING AND QUALITY SIGNALS

EXECUTION RECORDS

Per-run folders store prompts, agent outputs, action IDs, traces, decisions and response usage.

OBSERVED HISTORICAL ERRORS

One literal-expression release on v6; one unexplained escalation on v9. Both preserved.

HOSTED V11 COMMUNICATION SAFETY

Zero meter-fault confirmations and zero credit promises in five releases; no automatic retries.

Supervisor override rate and review-time reduction: NotMeasured. No actual human decision or provenanced baseline was available in the synthetic validation.

10 / DEPLOYMENT ARCHITECTURE

A tested path from prototype to service.

The hosted proof is synthetic; the enterprise boundary is designed to keep operational permissions explicit.

IMPLEMENTED AND TESTED

TYPED ADAPTER PORTS

Eight integration boundaries and synthetic adapters, with 138 passing contract checks.

HUMAN FINANCIAL CONTROL

Agent outputs cannot themselves authorize a credit or on-site inspection.

MIGRATION PROOF

Microsoft Agent Framework replay passed 11 of 11 local parity checks.

AUDITABLE FEEDBACK

Reviewer-only append records and explicit actual-versus-simulated review labels.

CONTROLLED PRODUCTION INTEGRATION PATH

01 / Governed sources

Billing, meter, policy and case APIs with access, freshness and provenance controls.

02 / Durable orchestration

Deploy Microsoft Agent Framework path ahead of Foundry Workflows preview retirement.

03 / Operational oversight

Human queue integration, restricted network/key access, telemetry, budgets and retention.

Scope: runtime-demonstrated multi-agent prototype and tested production integration architecture. These hosted runs used synthetic records; no live utility API or real supervisor disposition was represented as executed.

11 / EVIDENCE AND RELEASE

Inspectable by design.

The project is backed by original runs, per-case results, an auditable workflow and versioned criteria.

PUBLISHED SOURCE AND VALIDATION

GitHub / GridResolve-AI-MS-Agentathon

Control Center / Recorded and offline views

FOLLOW THE EVIDENCE**01 / Hosted runs**

evidence/runtime/ | eight recorded executions

02 / Acceptance

docs/SCENARIO_*_ACCEPTANCE_PLAN.md | per-case results

03 / Route records

docs/UPDATED_BRANCH_COVERAGE.md | actual action IDs

04 / Evaluation

docs/UPDATED_EVALUATION_METRICS.md | denominator-aware metrics

05 / Annotated trace

docs/ANNOTATED_TRACE_SYN-CASE-4003.md and 4001 trace

06 / Implementation

agents/ | tests/workflow_engine/ | integration/

Recorded status: v10 and v11 reference case 14/14 hosted acceptance; v11 no-follow-up case 12/12; five new cases executed. Unobserved hosted correction and non-approval routes remain classified as local proof only.